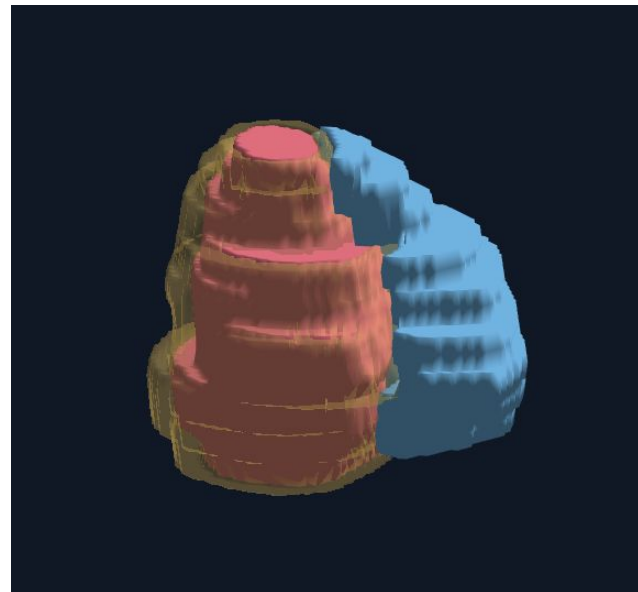
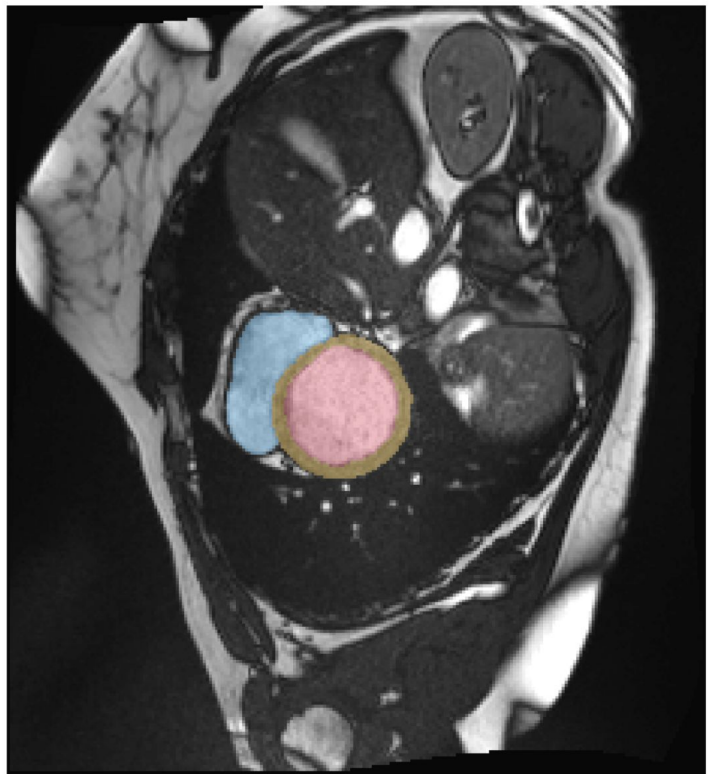


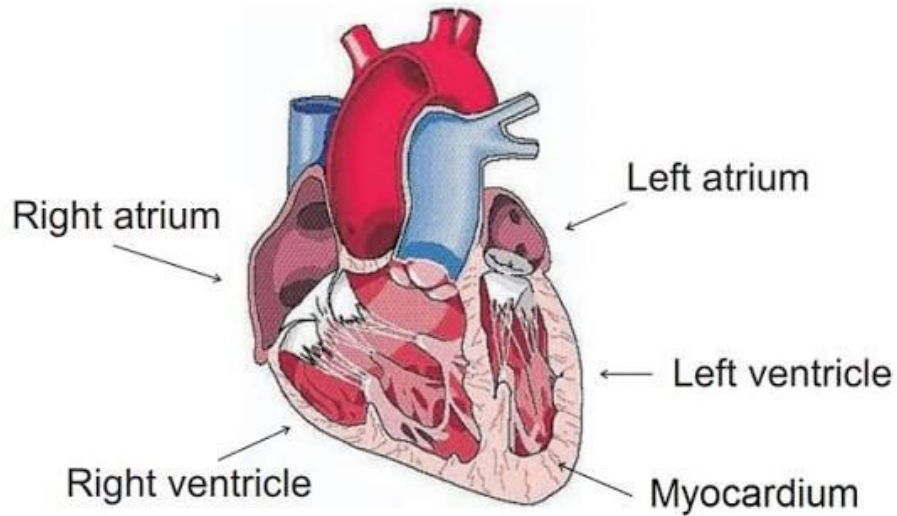
HeartFrame

From cardiac MRI to 3D anatomy and function.

What does my project do?



Background



Left Ventricle: The left ventricle is the thickest and strongest muscular chamber of the heart, responsible for pumping oxygen-rich blood out to the rest of the body

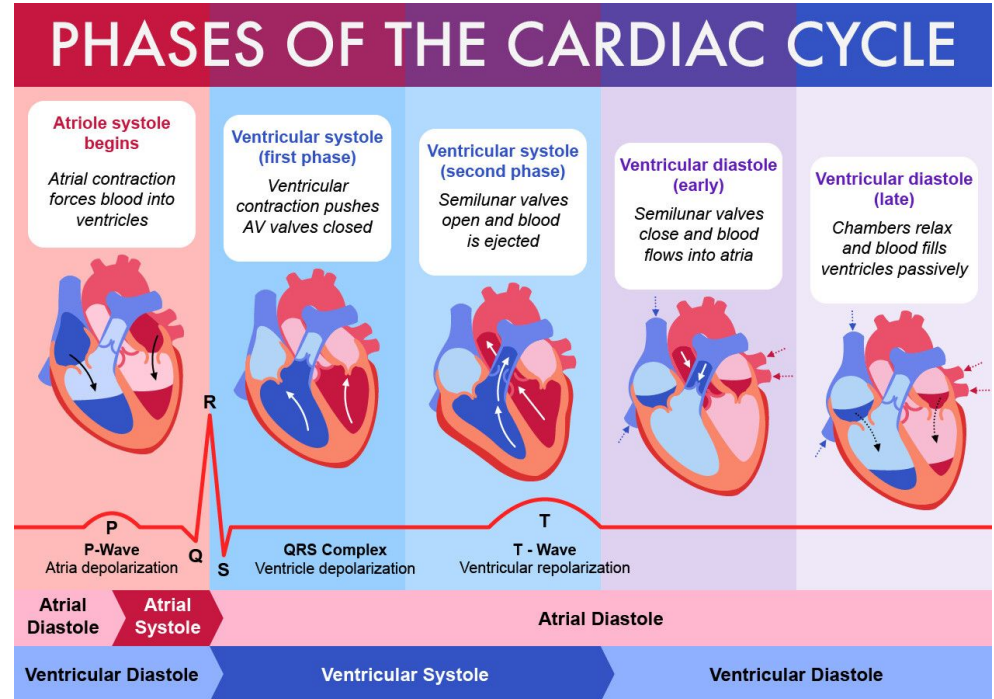
Right Ventricle: The right ventricle is the lower right chamber of the heart that receives oxygen-poor blood from the right atrium and pumps it to the lungs.

Myocardium: the thick, muscular middle layer of the heart wall that contracts to pump blood through the body

Phases of the Heart

Systole (Contraction): The heart muscle squeezes to push blood out of the chambers and into the body's arteries.

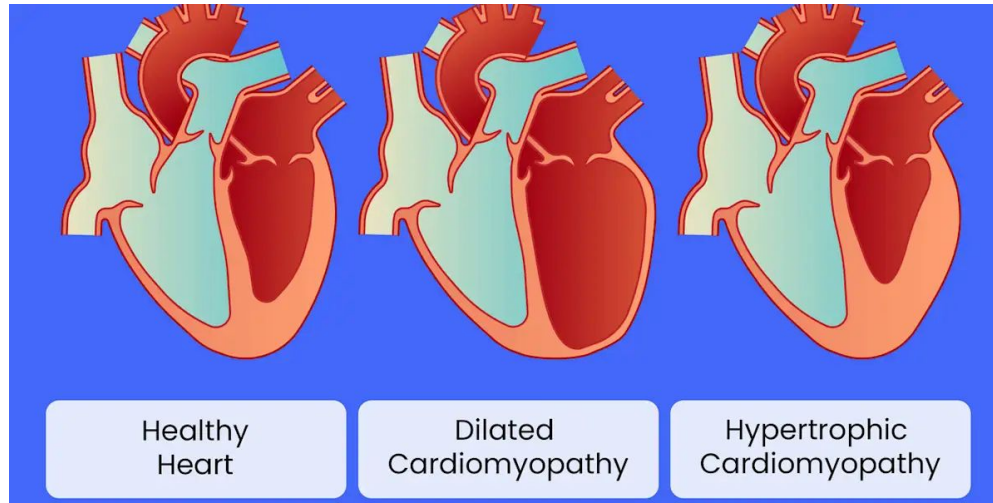
Diastole (Relaxation): The heart muscle relaxes and allows the chambers to refill with blood between beats.



What heart conditions can we see with this information?

Dilated cardiomyopathy (DCM) is a heart muscle disease where the main pumping chamber (left ventricle) stretches, thins, and grows larger, making it harder for the heart to pump blood.

Hypertrophic cardiomyopathy (HCM) is a genetic heart condition where the heart muscle becomes abnormally thick and stiff, making it harder for the heart to pump blood



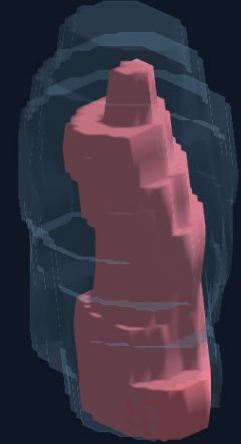
Normal



Dilated Cardiomyopathy



Hypertrophic Cardiomyopathy



What if we don't have expert labels?

- Trained a U-Net model on the 2D slices with labels
- Predicts where these three regions occur in the MRI Scan and builds a 3D model of the patient's heart
- Explores the impacts of failing model prediction on cardiac measurements, such as volume and ejection fraction across phases

Sources

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Appendix A: Dataset

Dataset: ACDC Cardiac MRI

- 150 patients across five groups: normal, prior heart attack, dilated cardiomyopathy, hypertrophic cardiomyopathy, and abnormal right ventricle.
- MRI slices at filling (ED) and contraction (ES) to visualize changes in ventricular anatomy.
- Expert masks of the LV, RV, and myocardium to train and evaluate the U-Net.
- Voxel spacing and phase identifiers to calculate chamber volumes, stroke volume, and ejection fraction.
- Patient group labels for comparing representative cardiac phenotypes.

Source: Bernard et al., IEEE TMI, 2018. DOI: [10.1109/TMI.2018.2837502](https://doi.org/10.1109/TMI.2018.2837502).

Deep Learning Techniques for Automatic MRI Cardiac Multi-structures Segmentation and Diagnosis: Is the Problem Solved?

Olivier Bernard, Alain Lalande, Clement Zotti, Frederick Cervenansky, Xin Yang, Pheng-Ann Heng, Irem Cetin, Karim Lekadir, Oscar Camara, Miguel Angel Gonzalez Ballester, Gerard Sanroma, Sandy Napel, Steffen Petersen, Georgios Tziritas, Elias Grinias, Mahendra Khened, Varghese Alex Kollerathu, Ganapathy Krishnamurthi, Marc-Michel Rohé, Xavier Pennec, Maxime Sermesant, Fabian Isensee, Paul Jäger, Klaus H. Maier-Hein, Peter M. Full, Ivo Wolf, Sandy Engelhardt, Chrisitan F. Baumgartner, Lisa M. Koch, Jelmer M. Wolterink, Ivana Išgum, Yeonggul Jang, Yoonmi Hong, Jay Patravali, Shubham Jain, Olivier Humbert, and Pierre-Marc Jodoin

Abstract—Delineation of the left ventricular cavity, myocardium and right ventricle from cardiac magnetic resonance images (multi-slice 2D cine MRI) is a common clinical task to establish diagnosis. The automation of the corresponding tasks has thus been the subject of intense research over the past decades. In this paper, we introduce the “Automatic Cardiac Diagnosis Challenge” dataset (ACDC), the largest publicly-available and fully-annotated dataset for the purpose of Cardiac MRI (CMR) assessment. The dataset contains data from 150 multi-equipments CMRI recordings with reference measurements and classification

from two medical experts. The overarching objective of this paper is to measure how far state-of-the-art deep learning methods can go at assessing CMRI, *i.e.* segmenting the myocardium and the two ventricles as well as classifying pathologies. In the wake of the 2017 MICCAI-ACDC challenge, we report results from deep learning methods provided by nine research groups for the segmentation task and four groups for the classification task. Results show that the best methods faithfully reproduce the expert analysis, leading to a mean value of 0.97 correlation score for the automatic extraction of clinical indices and an accuracy of 0.96 for automatic diagnosis. These results clearly open the door to highly-accurate and fully-automatic analysis of cardiac CMRI.

Appendix B: U-Net Model

U-Net fits a label for every pixel, with reasonably precise anatomical boundaries.

Its architecture has three useful components:

- Encoder: learns increasingly broad image features while reducing spatial resolution.
- Decoder: restores resolution to produce a segmentation map.
- Skip connections: carry detailed spatial features from the encoder to the decoder, helping locate boundaries.

U-Net: Convolutional Networks for Biomedical Image Segmentation

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Abstract. There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding-window convolutional network) on the ISBI challenge for segmentation of neuronal structures in electron microscopic stacks. Using the same network trained on transmitted light microscopy images (phase contrast and DIC) we won the ISBI cell tracking challenge 2015 in these categories by a large margin. Moreover, the network is fast. Segmentation of a 512x512 image takes less than a second on a recent GPU. The full implementation (based on Caffe) and the trained networks are available at <http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net>.

Appendix C: How I made the 3D visualizations

Uses scikit image marching cubes library to stack segmented MRI slices into a 3D volume, then convert the labeled boundaries into an interactive surface mesh. MRI voxel spacing preserves physical proportions.

Visualizations were made using plotly

Appendix D: Different Heart Metrics

Stroke volume is the amount of blood pumped out by the heart's ventricle during a single heartbeat.

Ejection fraction measures the percentage of blood pumped out of your heart's main pumping chamber, the left ventricle, with each heartbeat.

Cardiac Size vs Function

